

# Corentin Le Guen

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## Senior Backend Developer & Data Engineer

Software developer with experience in the banking industry, specialized in creating robust and scalable solutions for complex business needs. Passionate about leveraging technology to solve challenging problems and create value. Expertise in back-end development, real-time data processing, cloud deployment, and data analytics. Modernized architectures, developed critical applications, and implemented cutting-edge tools to support high-frequency trading and financial analysis. Combines technical proficiency with a strategic mindset to drive performance and innovation. Thrives in collaborative environments, delivering projects and mentoring others.

## Technologies

**Back-end:** Python, Java, Q/KDB+, Bash | **Databases:** Cassandra, SQLite3, MySQL, PostgreSQL, MongoDB, Redis  
**Testing:** Gatling, Mockito, Junit, Cucumber, Sonar | **Tools & Frameworks:** Flask, FastAPI, Splunk, DataDog, Git, Maven, GitLab / GitHub / Bitbucket, Postman, PyCharm, IntelliJ, Docker, Jenkins

## Professional Experience

### Morgan Stanley

March 2023 - March 2025

#### Data Scientist, Strategist & Data Analyst

Developed and maintained tools and applications for data manipulation, visualization, and reporting.

- Managed dashboard creation and maintenance for trading desks, providing key insights and performance metrics to facilitate informed decision-making.
- Collaborated with cross-functional teams to gather requirements, design data-driven solutions, and implement scalable data pipelines.
- Utilized KDB/Q for real-time data processing, querying & analysis to support high frequency trading strategies.
- Conducted comprehensive data audits and quality assessments to ensure data integrity and Reliability.
- Generated actionable insights through exploratory data analysis. Developed data-driven recommendations to optimize business processes and mitigate risks.

**Stack:** Q/KDB+ | Python | BitBucket | TeamCity

### FX Innovation

August 2021 - March 2023

#### Senior Java Developer & Tech Lead

**Client:** Desjardins. Tech lead on the modernization team. Java/Spring API for Oracle security. Deployment on Weblogic and Microsoft Azure. CI/CD on BitBucket / GitHub and Concourse. Testing with JUnit, Cucumber, Gherkin and Sonar. Project management with Jira and Confluence for documentation.

- Participated in modernizing the architecture.
- Updated and maintained an app to process files with Java/Spring.

**Stack:** Java | Spring | Swagger | Git | BitBucket | GitHub | Sonar | Junit | Cucumber | Gherkin | Concourse | Postman | Jira

### Systematix

September 2019 - August 2021

#### Senior Fullstack Developer

**Client:** National Bank of Canada. Maintained apps with NodeJS. Deployment with Ansible and Terraform. Tests with Axios, Mocha and Sonar. App monitoring with Splunk and DataDog. Installation and OpenText Exstream for document generation. CI/CD on BitBucket, Jenkins, Terraform on OpenShift.

- Updated Dockerfiles. Updated deployment scripts with VI on Linux servers.

**Stack:** NodeJS | FeathersJS | Express | Axios | Mocha | OpenShift | Jenkins | BitBucket | Ansible | Nexus | Vault | Jira | Confluence | Bash | Linux | VI | Docker | Dockerfile | Sonar | Splunk | DataDog | PostgreSQL | Postman | OpenText Exstream | Control-M | MFT

### Full stack Developer

**Client:** McGill University. Maintained a RabbitMQ app with Java and Junit. Added a security layer on API Connect.

- Created an algorithm and an app to confront JSON files with JQuery. Deployed on OpenShift and GitLab.

**Stack:** Java | RabbitMQ | Junit | JQuery | API Connect | NodeJS | Jira | Confluence

**Ippon Technologies, France**

**November 2018 - August 2019**

### Full stack Developer | 10 months

**Client:** BNP Paribas. Maintained a customer account consultation app on agile/scrum process. Testing with JUnit and Cucumber. Testing load on servers with Gatling. Helped two Java junior developer.

- Developed a Java, Spring app. Created producers and consumers Kafka. Upgraded Cassandra data models.

**Stack:** Java | Spring | JUnit | Cucumber | Gatling | Kafka | Cassandra | Solr | Git | Jira | Bash | Linux servers

**CGI, France**

**September 2018 - November 2018**

**Client:** SFR (Phone operator)

### Full stack Developer

Migrate ERP (Enterprise Resource Planning). Migrate Oracle servers. Maintained an app of professional accounts management. Upgraded a Java app (upgrade Java version from Java 6 to Java 8).

**Stack:** Java | Oracle | Git | Bash

**Altitude Infrastructure, France**

**July 2017 - August 2018**

### Developer

Application manager for phone operators. Maintained a C# app and developed a new C# app with .NET. Created a Front-end app to visualize KPI with C# and Bootstrap. Installed a new ERP (Enterprise Resource Planning) with transcoding tables. Migrated ERP from Adonix to SageX3. User testing.

**Stack:** C# | ASP.NET | Bootstrap 3 | JQuery | HTML | CSS | Bash | Git | Redmine | SQLServer

**Saagie, France**

**October 2016 - June 2017**

### Developer | Intern

Logging app to track customers journey on the website with Java, Hadoop, Cloudera and Hive. Logging javascript library with JQuery. UX/UI improvement.

**Stack:** Java | Kafka | Hive | Hadoop | Cloudera | Bash | Javascript | JQuery

## Personal Projects

### Personnal Website and portfolio

2025

A personnal blog / website using NodeJS, Sveltekit, TailwindCSS, Netlify and Flowbite

<https://cleguen.com>

### Project "Life Gameray"

2023

The Conway's Game of Life implementation using Raylib and Python.

<https://github.com/CorentinLeGuen/lifegameray>

### Project "Lotto 6.49 API"

2022

An API containing Lotto draws since 1982-06-12. Python3, Flask, SQLite, Docker

More here: <https://github.com/CorentinLeGuen/lotto-6-49-api>

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Project “Python on Page”</b>                                                                                                                                                                                                                                                     | 2022        |
| Running Python on browser using WebAssembly. Python3, Flask, WebAssembly<br><a href="https://github.com/CorentinLeGuen/python-on-page">https://github.com/CorentinLeGuen/python-on-page</a>                                                                                         |             |
| <b>Project “pycasso”</b>                                                                                                                                                                                                                                                            | 2022        |
| A project to transform a picture into ASCII code using Python and Numpy.<br><a href="https://github.com/CorentinLeGuen/pycasso">https://github.com/CorentinLeGuen/pycasso</a>                                                                                                       |             |
| <b>Renew project “Le Compte est Bon”</b>                                                                                                                                                                                                                                            | 2022        |
| Renew the project from 2013 with new technologies to solve the algorithm using Python<br>More here: <a href="https://github.com/CorentinLeGuen/le-compte-est-bon">https://github.com/CorentinLeGuen/le-compte-est-bon</a>                                                           |             |
| <b>Project “Scrap and analyse”</b>                                                                                                                                                                                                                                                  | 2021 & 2022 |
| A project to scrap data from websites and analyse them to predict next incoming update. Project hosted on a VPS (Virtual Proxy Server) on Linux. Python3, Pandas, BeautifulSoup4, Flask, Bash                                                                                       |             |
| <b>Projetct “Loto Quebec Last Result API”</b>                                                                                                                                                                                                                                       | 2021        |
| A project to scrap Data from Loto 6.49 website using Java<br><a href="https://github.com/CorentinLeGuen/LotoQuebec649LastResults">https://github.com/CorentinLeGuen/LotoQuebec649LastResults</a>                                                                                    |             |
| <b>Project “array list” in Go</b>                                                                                                                                                                                                                                                   | 2020        |
| A project made in Go to provide a Java arraylist implementation like. This project have been cloned more than 250 times and has 1 fork. Go. Code source here: <a href="https://github.com/CorentinLeGuen/le-compte-est-bon">https://github.com/CorentinLeGuen/le-compte-est-bon</a> |             |
| <b>Project “Simple KeyLogger”</b>                                                                                                                                                                                                                                                   | 2020        |
| A small project to save input keys into a log file. Python3<br><a href="https://github.com/CorentinLeGuen/SimpleKeyLogger">https://github.com/CorentinLeGuen/SimpleKeyLogger</a>                                                                                                    |             |
| <b>Project “Where is ISS?”</b>                                                                                                                                                                                                                                                      | 2018        |
| A front-end project to track the ISS position on earth globe. Javascript, JQuery, PlanetaryJS<br>Technology used: <a href="https://miniature.earth/demo/space-view.htm">https://miniature.earth/demo/space-view.htm</a>                                                             |             |
| <b>Project “Le Compte est Bon”</b>                                                                                                                                                                                                                                                  | 2013        |
| Created an algorithm to solve the game and a GUI with GTK. Made with Python2 and PyGTK.<br>More here: <a href="https://fr.wikipedia.org/wiki/Des_chiffres_et_des_lettres#Le_Compte_est_Bon">https://fr.wikipedia.org/wiki/Des_chiffres_et_des_lettres#Le_Compte_est_Bon</a>         |             |

## Education and Professional Development

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|---------------------------------------------------------------------------------------------------------------|------|
| <b>Data Science &amp; Machine Learning with Python</b> , Udemy Online course at udemy.com                     | 2022 |
| <b>Certificate of Language Training in English</b> , ACA Online course at lesateliers.ca                      | 2021 |
| <b>Specialization: Programming with Google Go</b> , Coursera Online course at coursera.org                    | 2019 |
| <b>Intermediate R Course</b> , DataCamp Online course at datacamp.com                                         | 2019 |
| <b>Deep Learning</b> , National Conservatory of “Arts et Métiers”, France Online course                       | 2018 |
| <b>Master degree of Computer Software Engineering</b> , University of Rouen Normandie, France                 | 2018 |
| <b>Licence Diploma of Development</b> , University of Rouen Normandie, France                                 | 2016 |
| <b>Introduction to C++ Development</b> , Polytechnic School of Lausanne, Swiss Online course at coursera.org  | 2013 |
| <b>Introduction to Java Development</b> , Polytechnic School of Lausanne, Swiss Online course at coursera.org | 2013 |

**Languages:** Bilingual in English and French